

Администрирование сетевых подсистем

Лабораторная работа № 9 — Настройка POP3/IMAP сервера
(Postfix + Dovecot)

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Цели и задачи работы

Цель лабораторной работы

Приобрести практические навыки установки и простейшего конфигурирования POP3/IMAP-сервера.

Реализовать доступ к почте по IMAP/POP3 и проверить работу сервиса с клиентской стороны.

Выполнение лабораторной работы

Разрешённые протоколы IMAP/POP3

```
[frank@server ~]$ sudo -i
[sudo] password for frank:
[root@server ~]# dnf -y install dovecot telnet
Extra Packages for Enterprise Linux 9 - x86_64 2.3 kB/s | 38 kB 00:16
Extra Packages for Enterprise Linux 9 - x86_64 430 kB/s | 20 MB 00:47
Rocky Linux 9 - BaseOS 392 B/s | 4.1 kB 00:10
Rocky Linux 9 - BaseOS 77 kB/s | 2.5 MB 00:33
Rocky Linux 9 - AppStream 191 B/s | 4.5 kB 00:24
Rocky Linux 9 - AppStream 227 kB/s | 9.5 MB 00:42
Rocky Linux 9 - Extras 270 B/s | 2.9 kB 00:11
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
dovecot x86_64 1:2.3.16-15.el9 appstream 4.7 M
telnet x86_64 1:0.17-85.el9 appstream 63 k
Installing dependencies:
clucene-core x86_64 2.3.3.4-42.20130812.e8e3d20git.el9 appstream 585 k
libexttextcat x86_64 3.4.5-11.el9 appstream 209 k
Transaction Summary
=====
Install 4 Packages

Total download size: 5.6 M
Installed size: 20 M
Downloading Packages:
(1/4): libexttextcat-3.4.5-11.el9.x86_64.rpm 40 kB/s | 209 kB 00:05
(2/4): clucene-core-2.3.3.4-42.20130812.e8e3d20 109 kB/s | 585 kB 00:05
(3/4): telnet-0.17-85.el9.x86_64.rpm 12 kB/s | 63 kB 00:05
(4/4): dovecot-2.3.16-15.el9.x86_64.rpm 3.2 MB/s | 4.7 MB 00:01
```

Проверка механизма аутентификации

```
Downloading Packages:
(1/4): libexttextcat-3.4.5-11.el9.x86_64.rpm    40 kB/s | 209 kB    00:05
(2/4): clucene-core-2.3.3.4-42.20130812.e8e3d20 109 kB/s | 585 kB    00:05
(3/4): telnet-0.17-85.el9.x86_64.rpm           12 kB/s | 63 kB     00:05
(4/4): dovecot-2.3.16-15.el9.x86_64.rpm        3.2 MB/s | 4.7 MB    00:01
-----
Total                                          452 kB/s | 5.6 MB    00:12
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                : 1/1
  Installing               : clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64 1/4
  Installing               : libexttextcat-3.4.5-11.el9.x86_64 2/4
  Running scriptlet: dovecot-1:2.3.16-15.el9.x86_64 3/4
  Installing           : dovecot-1:2.3.16-15.el9.x86_64 3/4
  Running scriptlet: dovecot-1:2.3.16-15.el9.x86_64 3/4
  Installing           : telnet-1:0.17-85.el9.x86_64 4/4
  Running scriptlet: dovecot-1:2.3.16-15.el9.x86_64 4/4
  Running scriptlet: telnet-1:0.17-85.el9.x86_64 4/4
  Verifying            : telnet-1:0.17-85.el9.x86_64 1/4
  Verifying            : libexttextcat-3.4.5-11.el9.x86_64 2/4
  Verifying            : clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64 3/4
  Verifying            : dovecot-1:2.3.16-15.el9.x86_64 4/4

Installed:
  clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64      dovecot-1:2.3.16-15.el9.x86_64
  libexttextcat-3.4.5-11.el9.x86_64                          telnet-1:0.17-85.el9.x86_64

Complete!
[root@server ~]#
```

Источники пользователей и паролей (PAM + passwd)

```
# protocol imap { }, local 127.0.0.1 { }, remote 10.0.0.0/8 { }

# Default values are shown for each setting, it's not required to uncomment
# those. These are exceptions to this though: No sections (e.g. namespace {})
# or plugin settings are added by default, they're listed only as examples.
# Paths are also just examples with the real defaults being based on configure
# options. The paths listed here are for configure --prefix=/usr
# --sysconfdir=/etc --localstatedir=/var

# Protocols we want to be serving.
protocols = imap pop3

# A comma separated list of IPs or hosts where to listen in for connections.
# "*" listens in all IPv4 interfaces, ":::" listens in all IPv6 interfaces.
# If you want to specify non-default ports or anything more complex,
# edit conf.d/master.conf.
#listen = *, :::

"/etc/dovecot/dovecot.conf" 101L, 4316B
```

24,21

To

Рис. 3: Настройка источников аутентификации в auth-system.conf.ext

Расположение почтовых ящиков (Maildir)

```
root@server:~  
# the auth service to run as root to be able to read this file.  
#auth_krb5_keytab =  
  
# Do NTLM and GSS-SPNEGO authentication using Samba's winbind daemon and  
# ntlm_auth helper. <doc/wiki/Authentication/Mechanisms/Winbind.txt>  
#auth_use_winbind = no  
  
# Path for Samba's ntlm_auth helper binary.  
#auth_winbind_helper_path = /usr/bin/ntlm_auth  
  
# Time to delay before replying to failed authentications.  
#auth_failure_delay = 2 secs  
  
# Require a valid SSL client certificate or the authentication fails.  
#auth_ssl_require_client_cert = no  
  
# Take the username from client's SSL certificate, using  
# X509_NAME_get_text_by_NID() which returns the subject's DN's  
# CommonName.  
#auth_ssl_username_from_cert = no  
  
# Space separated list of wanted authentication mechanisms:  
#  plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp  
#  gss-spnego  
# NOTE: See also disable_plaintext_auth setting.  
auth_mechanisms = plain  
  
##  
## Password and user databases  
##  
:wc
```


Межсетевой экран и запуск служб

```
##
# Location for users' mailboxes. The default is empty, which means that Dovecot
# tries to find the mailboxes automatically. This won't work if the user
# doesn't yet have any mail, so you should explicitly tell Dovecot the full
# location.
#
# If you're using mbox, giving a path to the INBOX file (eg. /var/mail/%u)
# isn't enough. You'll also need to tell Dovecot where the other mailboxes are
# kept. This is called the "root mail directory", and it must be the first
# path given in the mail_location setting.
#
# There are a few special variables you can use, eg.:
#
# %u - username
# %n - user part in user@domain, same as %u if there's no domain
# %d - domain part in user@domain, empty if there's no domain
# %h - home directory
#
# See doc/wiki/Variables.txt for full list. Some examples:
#
# mail_location = maildir:~/Maildir
# mail_location = mbox:~/mail:INBOX=/var/mail/%u
# mail_location = mbox:/var/mail/%d/%n/INBOX=/var/indexes/%d/%n/INBOX
#
# <doc/wiki/MailLocation.txt>
#
# mail_location = maildir:~/Maildir
#
# If you need to set multiple mailbox locations or want to change default
# namespace settings, you can do it by defining namespace sections.
#
# You can have private, shared and public namespaces. Private namespaces
# are for user's personal mails. Shared namespaces are for accessing other
# users' mailboxes that have been shared. Public namespaces are for shared
# mailboxes that are managed by sysadmin. If you create any shared or public
# namespaces you'll typically want to enable ACL plugin also, otherwise all
# users can access all the shared mailboxes, assuming they have permissions
# on filesystem level to do so.
```

```
[root@server ~]#  
[root@server ~]# postconf -e 'home_mailbox = Maildir/'  
[root@server ~]#  
[root@server ~]#
```

Рис. 6: Записи в журнале Postfix и Dovecot

Проверка почтового ящика на сервере

```
[root@server ~]#  
[root@server ~]# postconf -e 'home_mailbox = Maildir/'  
[root@server ~]#  
[root@server ~]#  
[root@server ~]# firewall-cmd --get-services  
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp2  
bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet  
et bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collectd cond  
or-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls dock  
er-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger foreman foreman-proxy fr  
eeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master  
git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp ipp-client ipsec irc ircs is  
csi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver  
kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-secure kube-nodep  
ort-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap lda  
ps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcac  
he mindlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-ns netdata-dashboar  
d nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pncd pmproxy  
pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3ne  
tsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt  
-master samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmpv1 snmpv2-trap sm  
ptrap spideroak-lansync spotify-sync squid sssd ssh steam-streaming svdrp svn syncthing syncthing-gui syncthin  
g-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp-client v  
dsm vnc-server warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-  
discovery-udp wsman wsmans xdmpc xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerot  
ier  
[root@server ~]# firewall-cmd --add-service=pop3 --permanent  
success  
[root@server ~]# firewall-cmd --add-service=pop3s --permanent  
success  
[root@server ~]# firewall-cmd --add-service=imap --permanen  
success  
[root@server ~]# firewall-cmd --add-service=imaps --permanent  
bash: firewall-cmd: command not found...  
^[[A^[[B^[[B  
^C  
[root@server ~]# firewall-cmd --add-service=imaps --permanent  
success  
[root@server ~]# firewall-cmd --reload  
success  
[root@server ~]# firewall-cmd --list-services  
cockpit dhcpv6-client imap imaps pop3 pop3s smtp ssh  
[root@server ~]#
```

Evolution: идентификация пользователя

```
[root@server ~]#  
[root@server ~]# restorecon -vR /etc  
[root@server ~]# systemctl restart postfix  
[root@server ~]# systemctl enable dovecot  
Created symlink /etc/systemd/system/multi-user.target.wants/dovecot.service → /usr/lib/systemd/system/dovecot.service.  
[root@server ~]# systemctl start dovecot  
[root@server ~]#
```

Рис. 8: Настройка учетной записи в Evolution — идентификация

Evolution: входящая почта (IMAP)

```
[root@server ~]# MAIL=~/.Maildir mail
bash: mail: command not found...
[root@server ~]# MAIL=~/.Maildir mail
s-nail: No mail for root at /root/.Maildir
s-nail: /root/.Maildir: No such entry, file or directory
[root@server ~]#
[root@server ~]# dovecadm mailbox list -u user
INBOX
[root@server ~]# dovecadm mailbox list -u frank
INBOX
[root@server ~]#
```

Рис. 9: Настройка входящей почты IMAP в Evolution

Evolution: исходящая почта (SMTP)

Проверены параметры SMTP: сервер mail.user.net, порт 25, без шифрования и без аутентификации.

The image shows two overlapping windows from the Evolution mail client. The top window is titled 'Identity' and contains a sidebar with links: 'Welcome', 'Restore from Backup', 'Identity' (selected), 'Receiving Email', 'Sending Email', 'Account Summary', and 'Done'. The main area of this window has a heading 'Please enter your name and email address below. The "optional" fields below do not need to be filled in, unless you wish to include this information in email you send.' Below this is a section for 'Required Information' with fields for 'Full Name' (containing 'frank') and 'Email Address' (containing 'frank@frank.net'). There is also an 'Optional Information' section with fields for 'Reply-To', 'Organization', and 'Aliases' (with an 'Add' button). The bottom window is titled 'Receiving Email' and has a similar sidebar. Its main area shows 'Server Type' set to 'IMAP' with a description 'For reading and storing mail on IMAP servers.' Below is a 'Configuration' section with fields for 'Server' (mail.frank.net), 'Port' (143), and 'Username' (frank). At the bottom is a 'Security' section with 'Encryption method' set to 'STARTTLS after connecting'.

Welcome
Restore from Backup
Identity
Receiving Email
Sending Email
Account Summary
Done

Please enter your name and email address below. The "optional" fields below do not need to be filled in, unless you wish to include this information in email you send.

Required Information

Full Name: frank

Email Address: frank@frank.net

Optional Information

Reply-To:

Organization:

Aliases: Add

Receiving Email

Welcome
Restore from Backup
Identity
Receiving Email
Receiving Options
Sending Email
Account Summary
Done

Server Type: IMAP

Description: For reading and storing mail on IMAP servers.

Configuration

Server: mail.frank.net Port: 143

Username: frank

Security

Encryption method: STARTTLS after connecting

Отправка и получение тестовых писем

Receiving Options

Welcome

Restore from Backup

Identity

Receiving Email

Receiving Options

Sending Email

Account Summary

Done

Checking for New Mail

☒ Check for new messages every 60 - + minutes

☐ Check for new messages in all folders

☐ Check for new messages in subscribed folders

☐ Use Quick Resync if the server supports it

☒ Listen for server change notifications

Connection to Server

Number of concurrent connections to use 3 - +

☒ Enable full folder update on metered network

Folders

☒ Show only subscribed folders

Options

☐ Apply filters to new messages in all folders

☒ Apply filters to new messages in Inbox on this server

☐ Check new messages for Junk contents

☐ Only check for Junk messages in the Inbox folder

☐ Synchronize remote mail locally in all folders

☐ Do not synchronize locally mails older than 1 - + years

Cancel Back Next

Проверка POP3 через Telnet

Sending Email

Welcome

Restore from Backup

Identity

Receiving Email

Receiving Options

Sending Email

Account Summary

Done

Server Type: SMTP

Description: For delivering mail by connecting to a remote mailhub using SMTP.

Configuration

Server: mail.frank.net

Port: 25

☐ Server requires authentication

Security

Encryption method: TLS on a dedicated port

Authentication

Type: Check for Supported Types PLAIN

Username: frank

Account Summary

Welcome

Restore from Backup

Identity

Receiving Email

Receiving Options

Sending Email

Account Summary

Done

This is a summary of the settings which will be used to access your mail.

Account Information

Name: frank@frank.net

The above name will be used to identify this account.
Use for example, "Work" or "Personal".

Personal Details

Full Name: frank

Email Address: frank@frank.net

Receiving	Sending
Server Type: imap	smtp
Server: mail.frank.net	mail.frank.net
Username: frank	frank
Security: STARTTLS	TLS

Выводы по проделанной работе

В ходе лабораторной работы был установлен и настроен почтовый сервер на базе Postfix и Dovecot.

Выполнена настройка протоколов IMAP и POP3, обеспечено хранение почты в формате Maildir и корректная аутентификация пользователей.

Проверена работа почтовой системы с использованием Evolution, утилит mail/doveadm и проверки POP3 через Telnet.

Подтверждена корректная доставка, получение и удаление сообщений, а также готовность конфигурации к автоматическому развёртыванию в Vagrant.